

UNIVERSITY DATABASE DATA DICTIONARY

1. ENTITY DESCRIPTIONS

Entity Name	Description
Student	Stores personal and academic information for every student registered in the university.
Instructor	Stores personal and professional details of university faculty members.
Major	Represents the academic departments or fields of study.
Course	A catalog of all courses offered by the university.
Room	Physical locations (classrooms, labs, halls) available for scheduling classes.
Schedule	The operational timetable linking a specific course to a room, instructor, and time.
Enrollment	A transaction table recording which student is taking which course, including grades.
Std_Contact	Normalized table for storing multiple phone numbers for students.
Inst_Contact	Normalized table for storing multiple phone numbers for instructors.
CourseMajor	Junction table defining which courses belong to which major.
InstructorMajor	Junction table defining which instructors belong to which major.

2. ATTRIBUTE DEFINITIONS

Table: Student

Attribute	Type	Key	Req	Description
id	INT	PK	Yes	Unique identifier for the student.
ssn	INT	UQ	Yes	Social Security Number (Unique).
fname	VARCHAR(10)		Yes	First Name.
lname	VARCHAR(10)		No	Last Name.
address	VARCHAR(100)		No	Residential address.
sex	VARCHAR(1)		No	Gender (M/F).
date of birth	DATE		No	Birthdate.
level	VARCHAR(10)		No	Academic year.
gpa	REAL		No	Grade Point Average (0.0 - 4.0).
status	VARCHAR(15)		No	Current state (Active, Probation, Warning).
majorID	INT	FK	Yes	Links to Major table.

Table: Instructor

Attribute	Type	Key	Req	Description
id	INT	PK	Yes	Unique identifier for the instructor.
ssn	INT	UQ	Yes	Social Security Number.
fname	VARCHAR(10)		Yes	First Name.
lname	VARCHAR(10)		No	Last Name.
title	VARCHAR(20)		No	Job Title (Professor, Lecturer, etc.).
salary	INT		No	Monthly or annual salary.

Table: Major

Attribute	Type	Key	Req	Description
id	INT	PK	Yes	Unique identifier for the Major.
name	VARCHAR(50)		No	Name of the Major.
deanID	INT	FK/UQ	Yes	Links to Instructor (Head of Department).

Table: Schedule

Attribute	Type	Key	Req	Description
scheduleID	INT	PK	Yes	Unique ID for the schedule entry.
roomNum	INT	FK	Yes	Where the class is held.
courseCode	VARCHAR(20)	FK	Yes	What is being taught.
instID	INT	FK	Yes	Who is teaching.
time	TIME		No	Start time.
day	VARCHAR(3)		No	Day of week.

Table: Enrollment

Attribute	Type	Key	Req	Description
id	INT	PK	Yes	Unique Transaction ID.
stdID	INT	FK	Yes	Links to Student.
courseCode	VARCHAR(20)	FK	Yes	Links to Course.
grade	VARCHAR(2)		No	Final Grade.
status	VARCHAR(15)		No	Enrollment state.

Table: Course

Attribute	Type	Key	Req	Description
code	VARCHAR(20)	PK	Yes	Unique Course Code.
name	VARCHAR(50)		No	Full name of the course.
credits	INT		No	Credit hours.

Table: Room

Attribute	Type	Key	Req	Description
num	INT	PK	Yes	Room Number.
floor	INT		No	Floor number.
bulding	INT		No	Building identifier.
capacity	INT		No	Max capacity.
Type	VARCHAR(50)		No	Room type.

3. RELATIONSHIP DETAILS

Relationship	Type	Description
Student - Major	One-to-Many	A Major has many students, but a Student belongs to only one Major (via MajorID).
Instructor - Major (Dean)	One-to-One	A Major has one Dean, and a Dean manages one Major.
Student - Course	Many-to-Many	Resolved via Enrollment table.
Instructor - Course	Many-to-Many	Resolved via Schedule table.
Course - Major	Many-to-Many	Resolved via CourseMajor table.
Instructor - Major (Work)	Many-to-Many	Resolved via InstructorMajor table.
Room - Schedule	One-to-Many	A Room can host many different scheduled classes.
Student - Contact	One-to-Many	A Student can have multiple phone numbers.
Instructor - Contact	One-to-Many	An Instructor can have multiple phone numbers.